NBS4MED self-evaluation report

Generated: 2026-10-17

Demo version: illustrative estimates, to be reviewed when project data become available

Preliminary prototype results for the area as it is today. The methodology will be reviewed against the PROMEA framework and the NBS repository. The scores are NBS4MED prototype scores and are not an official score or certification of any external organisation or standard.

1. Assessment area

Assessment area	İzmir Green Roof Pilot
Country	Türkiye
City / municipality	İzmir
Assessment scale	Site / Project Area
Organisation	Not stated
Assessment date	2026-10-17

2. Assessment of the five criteria (provisional)

The assessment describes existing NBS integration (C1) and the enabling and implementation conditions (C2 to C5). It is provisional and separate from the suitability, impact and cost of a solution.

Completeness

14 of 14 indicators have a numeric score. 0 Unknown. 0 Not applicable.

Criterion profile

Current NBS integration (C1). How far NBS are already integrated in the area today. This criterion does not measure environmental effectiveness, such as cooling or flood reduction.

Criterion	Score	Level	Completeness and confidence
C1 NBS Integration Status	3 / 6 (50%)	Partial	3 of 3 indicators scored

Enabling and implementation conditions (C2 to C5). Policy, capacity, engagement and funding that support planning, delivery and long-term upkeep of NBS.

Criterion	Score	Level	Completeness and confidence
C2 Policy & Planning	3 / 6 (50%)	Partial	3 of 3 indicators scored
C3 Institutional & Technical Capacity	4 / 6 (67%)	Adequate	3 of 3 indicators scored
C4 Stakeholder Engagement	1 / 4 (25%)	Insufficient	2 of 2 indicators scored
C5 Funding & Sustainability	3 / 6 (50%)	Partial	3 of 3 indicators scored

Provisional combined assessment score (secondary; not a validated measure of NBS integration)

14 / 28 (50%) **Partial**

The overall score gives every indicator equal weight, so a criterion with more indicators counts for more. Scored indicators per criterion: C1 3, C2 3, C3 3, C4 2, C5 3. Provisional project thresholds, not taken from any external standard: Insufficient below 30%, Partial 30 to 54%,

Adequate 55 to 79%, Strong 80% and above. Project partners will review them.

Gaps and strengths by indicator

Priority gaps (score 0)	1.3 Link to grey infrastructure: NBS are planned separately from drainage, roads and buildings, or not linked to them at all. 2.3 Targets and monitoring: No measurable NBS targets exist. 4.1 Community participation: Residents are not involved in planning or designing NBS. 5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS.
Developing (score 1)	1.2 Coverage of public space: Some public space has green or blue features, but coverage is partial or uneven, or falls short of a local target where one exists. 2.2 Regulation and standards: Some rules or standards encourage NBS, but they are voluntary, partial or rarely applied. 3.1 Dedicated staff or unit: Staff handle NBS alongside other duties, or responsibility is limited to one department. 3.3 Data and monitoring tools: Some data exist, but they are incomplete, out of date or only occasionally used in decisions. 4.2 Coordination across departments and owners: Coordination happens case by case, or only between some of the actors. 5.3 Co-benefits valued in decisions: Co-benefits are mentioned in funding decisions, but are not measured or valued.
Strengths (score 2)	1.1 NBS inventory 2.1 NBS in strategic plans 3.2 Technical know-how 5.1 Capital funding
Information gaps (Unknown)	–
Not applicable	–

Priority gaps and improvement actions

Shown in a provisional display order: answers scored 0 first, then 1, then Unknown, each in indicator order. This is not a validated ranking of impact or urgency. The actions are general prompts based on your answers, not findings about the site.

#	Indicator	Type	Finding	Why it matters	Suggested action	Evidence needed
1	1.3 — Link to grey infrastructure	Gap	NBS are planned separately from drainage, roads and buildings, or not linked to them at all.	NBS planned with drainage, roads and buildings are cheaper to build, easier to maintain and more likely to work as intended.	Introduce a step in infrastructure projects to consider NBS at the design stage.	Project design documents, drainage or road plans and procedures for infrastructure projects.
2	2.3 — Targets and monitoring	Gap	No measurable NBS targets exist.	Measurable targets and regular reporting show whether NBS are progressing and allow plans to be corrected.	Define a small set of measurable NBS targets, for example area, number or condition of solutions, with a baseline year.	Target statements, monitoring indicators and progress reports.
3	4.1 — Community participation	Gap	Residents are not involved in planning or designing NBS.	Early participation improves fit with local needs and support for upkeep.	Introduce a participation step, such as workshops, in the early stage of NBS planning.	Workshop or consultation records, and records of how input changed decisions.
4	5.2 — Long-term maintenance budget	Gap	No budget covers maintenance, watering or replacement of NBS.	NBS need continuing care. Without it they can fail within a few years.	Prepare an operation and maintenance budget and clarify who is responsible for funding it.	Maintenance plans, annual budget lines and financial documentation.

#	Indicator	Type	Finding	Why it matters	Suggested action	Evidence needed
5	1.2 — Coverage of public space	Partial	Some public space has green or blue features, but coverage is partial or uneven, or falls short of a local target where one exists.	Coverage shows how much of the public space can already contribute to cooling, shade and water management.	Identify the districts with low or uneven coverage and compare coverage with a local target, or agree on one.	Land-cover or green-space maps, municipal green-area statistics and any local target.
6	2.2 — Regulation and standards	Partial	Some rules or standards encourage NBS, but they are voluntary, partial or rarely applied.	Rules and incentives make NBS a normal part of development, not a voluntary extra.	Clarify and apply the existing rules or incentives more consistently, and track how often they are applied.	Zoning regulations, building codes, green-area standards and examples of their application.
7	3.1 — Dedicated staff or unit	Partial	Staff handle NBS alongside other duties, or responsibility is limited to one department.	Clear responsibility keeps NBS on the agenda and avoids tasks falling between departments.	Formalise the role and extend its mandate across departments.	Organisation charts, job descriptions or a decision naming the responsible unit.
8	3.3 — Data and monitoring tools	Partial	Some data exist, but they are incomplete, out of date or only occasionally used in decisions.	Data are needed to place NBS where they help most and to check whether they work.	Complete and update the datasets and agree how they feed into decisions and monitoring.	Dataset lists, data-sharing agreements and examples of data used in decisions.
9	4.2 — Coordination across departments and owners	Partial	Coordination happens case by case, or only between some of the actors.	Many NBS depend on land, assets or budgets held by different actors.	Turn case-by-case coordination into a regular mechanism, such as a standing working group.	Meeting records, agreements between actors or a working-group mandate.
10	5.3 — Co-benefits valued in decisions	Partial	Co-benefits are mentioned in funding decisions, but are not measured or valued.	Counting co-benefits can strengthen the case for funding NBS compared with grey alternatives.	Measure or value the main co-benefits and use them in funding decisions.	Funding proposals, business cases and any co-benefit studies.

Confidence in the answers

Of the 14 scored indicators: 1 documented evidence, 0 staff estimate, 0 informal judgement, 13 confidence not stated. Confidence is reported next to the answers. It does not change any score.

Answers in detail

ID	Indicator	Answer	Rationale	Evidence or source	Confidence
1.1	NBS inventory	2 of 2	Based on the municipal GIS layer	Inventory report 2024	High: documented evidence
1.2	Coverage of public space	1 of 2	–	–	Not stated
1.3	Link to grey infrastructure	0 of 2	–	–	Not stated
2.1	NBS in strategic plans	2 of 2	–	–	Not stated
2.2	Regulation and standards	1 of 2	–	–	Not stated
2.3	Targets and monitoring	0 of 2	–	–	Not stated
3.1	Dedicated staff or unit	1 of 2	–	–	Not stated
3.2	Technical know-how	2 of 2	–	–	Not stated
3.3	Data and monitoring tools	1 of 2	–	–	Not stated
4.1	Community participation	0 of 2	–	–	Not stated
4.2	Coordination across departments and owners	1 of 2	–	–	Not stated
5.1	Capital funding	2 of 2	–	–	Not stated
5.2	Long-term maintenance budget	0 of 2	–	–	Not stated
5.3	Co-benefits valued in decisions	1 of 2	–	–	Not stated

Is there a local target for NBS or for green and blue space coverage? No. Is the current coverage known (mapped or measured)? No.

Site context (not scored)

Urban Density	High
Public Land Available for NBS	Limited
Local Climate Risks	Heatwaves; Flooding; Drought

How the assessment scores are calculated (provisional)

- Each indicator is scored 0, 1 or 2 using the provisional project rubric shown in the assessment.
- Unknown is missing information and is never counted as 0. Not applicable (with a justification) is also left out. Both are reported separately.
- A criterion percentage is the sum of its scored indicators divided by 2 times the number of scored indicators. It is shown only when the criterion has no Unknown answer. A criterion whose indicators are all Not applicable is shown as Not applicable.
- The overall percentage uses the same rule across all indicators and is shown only when no indicator is Unknown. With every indicator scored, it is the total divided by 28. Every indicator has equal weight, so criteria with more indicators weigh more.
- Levels use provisional project thresholds: Insufficient below 30%, Partial 30 to 54%, Adequate 55 to 79%, Strong 80% and above.
- The scores are not multiplied by confidence. Confidence is reported separately, indicator by indicator.
- C1 describes current NBS integration only. It is not a measure of environmental effectiveness.

3. Applicability screening

Screening only, not engineering approval. These checks do not change the assessment score or the match score. Answers about planting space, soil and water are shared by the planting-based options.

No solution has been checked yet. Applicability statuses show Not yet checked.

4. Candidate solutions and reasons

Options are listed in the order of a provisional rule-based match to your selected risks and site context. This is not a validated ranking. Applicability statuses come from optional checks and are not engineering approval. The match is a provisional rule-based description of how well an option addresses the selected risks and the land context. It is not a probability of success, a feasibility assessment or a predicted climate impact.

#	Option	Category (provisional)	Rule-based match score	Screening status	Why it matches the selected risks	To verify	Implementation conditions
1	Green roof	Green infrastructure	56 / 100	Not yet checked	Heatwaves (moderately); Flooding (moderately); Drought (weakly). Uses existing roof space, not public land. Roof availability and structure still need checking.	No check has been answered yet. The recommendation is preliminary.	5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS.
2	Green corridor	Green infrastructure	44 / 100	Not yet checked	Heatwaves (strongly); Flooding (weakly). Not a target of this option: Drought. Land need fits the available public land.	No check has been answered yet. The recommendation is preliminary.	5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS.
3	Permeable paving	Blue-green water management · Hybrid/integrated solutions	44 / 100	Not yet checked	Flooding (strongly); Drought (weakly). Not a target of this option: Heatwaves. Land need fits the available public land.	No check has been answered yet. The recommendation is preliminary.	5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS.
4	Urban forest (Miyawaki method)	Green infrastructure · Ecosystem restoration	41 / 100	Not yet checked	Heatwaves (strongly); Flooding (weakly); Drought (weakly). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS.
5	Rain garden and bioswale	Blue-green water management	41 / 100	Not yet checked	Heatwaves (weakly); Flooding (strongly); Drought (weakly). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS.

#	Option	Category (provisional)	Rule-based match score	Screening status	Why it matches the selected risks	To verify	Implementation conditions
6	Smart eco-park	Hybrid/integrated solutions · Green infrastructure	38 / 100	Not yet checked	Heatwaves (strongly); Flooding (moderately); Drought (moderately). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS.
7	Rainwater retention basin	Blue-green water management · Hybrid/integrated solutions	38 / 100	Not yet checked	Heatwaves (weakly); Flooding (strongly); Drought (strongly). Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS.
8	Pocket park	Green infrastructure	18 / 100	Not yet checked	Heatwaves (moderately); Flooding (weakly). Not a target of this option: Drought. Land need exceeds the available public land, which lowers the provisional match.	No check has been answered yet. The recommendation is preliminary.	5.2 Long-term maintenance budget: No budget covers maintenance, watering or replacement of NBS.

Excluded and other options

These options have a rule-based match score of 0, so they are not recommended for the selected risks and context. A score of 0 is not a finding that an option is technically unsuitable.

Option	Category (provisional)	Screening status	Why it matches the selected risks
Vegetated dune and shoreline	Ecosystem restoration	Other options	Its land need exceeds the public land available, which brings the match score to 0.

How the rule-based match score is calculated (provisional)

- Each option has a weight from 0 (not a target) to 3 (strong) for heatwaves, flooding and drought. These are the earlier demo weights, kept as provisional rules pending methodology review. They have not been validated by PROMEA.
- Matching rules for Coastal Storms and Land Degradation have not been defined. When either is selected, or when Not Sure is selected, no match score or ranking is produced.
- Score = sum of the weights for your selected risks, divided by 3 times the number of selected risks, times 100.
- Land context: 15 points are subtracted for each level by which the option's land need exceeds the public land available, and 10 more points when urban density is high and the land need is high. The score is limited to 0 to

100.

- Urban density and public land availability are qualitative answers, not numerical thresholds. If public land availability is Unknown, no match score is produced. Institutional gaps (funding, maintenance, technical know-how) are not used to change the score. They are shown as implementation conditions.
- Climate type is not yet used in matching. It will be considered once verified location-specific information and decision rules are available.
- Public land availability is not used as evidence that a roof is available. Roofs are checked separately in the applicability checks.
- Screening: an option is Not applicable only when a basic condition is answered No (for example no roof, no coast, no runoff source). Unknown, or a No on a condition that can be assessed or designed (for example structural suitability), gives Requires verification.

Option	Heatwaves	Flooding	Drought
Urban forest (Miyawaki method)	3	1	1
Green roof	2	2	1
Green corridor	3	1	0
Rain garden and bioswale	1	3	1
Permeable paving	0	3	1
Pocket park	2	1	0
Smart eco-park	3	2	2
Rainwater retention basin	1	3	3
Vegetated dune and shoreline	0	1	0

Working alignment with the ten PROMEA considerations

Working framework used by the project team. This is not a formal PROMEA validation. Items marked Pending are not yet operational in this prototype.

Consideration	Status	Detail
Climate Risks	Partially supported	Selection of five risks (multiple allowed, plus Not Sure) is implemented. Provisional matching rules exist only for heatwaves, flooding and drought.
Climate Type	Pending further development	Not asked and not inferred. Needs verified location-specific information and decision rules.
Urban Context	Partially supported	Urban density and public land availability are asked as qualitative answers and used in the provisional matching.
NBS Categories	Partially supported	Provisional category labels on each option. They do not affect matching.
Cost	Partially supported	Indicative demo cost ranges for five of the nine options, not local prices.
Stakeholders	Partially supported	C4 Stakeholder Engagement (4.1, 4.2) on the provisional 0 to 2 rubric. Not used in matching.
Regulations	Partially supported	C2 Policy & Planning (2.1 to 2.3) on the provisional 0 to 2 rubric. Not used in matching.
Technical Expertise	Partially supported	C3 Institutional & Technical Capacity (3.1 to 3.3). Technical know-how (3.2) is also shown as an implementation condition.
Maintenance	Partially supported	Indicator 5.2, shown as an implementation condition, and an annual maintenance cost range for supported options.

Consideration	Status	Detail
Evaluation	**Pending further development**	Indicators 2.3 and 3.3 are assessed, but there is no monitoring or evaluation module for NBS performance.

5. Impact and cost exploration (illustrative)

The impact and cost step was not completed. Only the assessment is reported.

Demo version: illustrative estimates, to be reviewed when project data become available